IoT-Enabled Smart bus route monitoring and revenue management system

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Public bus transport operations face multiple difficulties which include coordinating their services and executing their trips on time while experiencing problems with real-time bus tracking and driver behavior monitoring during daily operations. Transport stakeholders frequently experience operational challenges because they cannot access information about bus availability and trip status and breakdown events and unsafe or inefficient driving behavior. The existing limitations decrease service reliability while creating obstacles for effective management of operational interruptions. The development of a mobile-based smart solution requires creation of an operational reporting system which enables real-time driver monitoring.

The research component focuses on creating an IoT-based mobile application for smart city bus operations which designates bus drivers as main users and bus owners as secondary users. The mobile application enables operational functions which include bus availability updates and trip status reporting and breakdown notification. The proposed solution uses onboard IoT sensor data which includes longitude and current speed and acceleration and gyroscope values and timestamp data to monitor driving behavior in actual time..

The system applies rule-based detection mechanisms to identify important driving events which include overspeed and abnormal slow movement and aggressive braking and aggressive turning. The mobile application delivers alerts and warnings to drivers when the system detects such events because this feature helps drivers become aware of dangers while they operate their vehicles. The application provides operational summaries together with incident visibility which different stakeholders can access. The component integrates operational reporting with sensor-based driver monitoring to create an effective driver-side interface which enhances the complete smart bus management system. The expected outcome is improved operational visibility together with better driver awareness and enhanced efficiency and safety in city bus transport services.

Keywords: Driver Monitoring, Mobile Application, IoT Sensors, Bus Operations, Aggressive Driving Detection, Smart Transport Systems.

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LIST OF ABBREVIATIONS

Abbreviation	Description
API	Application Programming Interface
DB	Database
ETA	Estimated Time of Arrival
GPS	Global Positioning System
IMU	Inertial Measurement Unit
IoT	Internet of Things
ITS	Intelligent Transportation System
REST	Representative State Transfer
UI	User Interface
RPTA	Road Passenger Transport Authority

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1. INTRODUCTION

1.1 Background and Context

Public bus systems in both urban and suburban regions serve as essential transportation services that enable numerous passengers to travel throughout the day. Buses in Sri Lanka must follow approved routes and schedules and service conditions which transport authorities have established for both city and suburban bus operations. Daily bus operations face multiple challenges because of delays which occur during breakdowns and because bus service availability is inconsistent and stakeholders lack access to real-time information. The two issues lead to decreased service reliability which results in operational difficulties for both bus operators and transport authorities. Current bus operations face major challenges because drivers and bus owners cannot use digital solutions to report their operational status during service. Information about bus service availability and scheduled trip operations and breakdowns fails to reach its intended audience through efficient communication methods. Stakeholders face challenges in managing operations because they lack operational information which causes service interruptions to take longer to resolve and hinders their ability to deliver continuous service..

Driver conduct throughout service delivery operations together with operational reporting problems presents a major challenge which reduces both service standards and road safety. Driving patterns which include overspeeding and aggressive braking and harsh turning and abnormal slow movement create negative effects which impact passenger comfort and operational efficiency and safety. The traditional bus operation systems lack capabilities to monitor driver behavior through real-time tracking and to supply drivers with instant feedback while they work. The development of an intelligent mobile system needs to take both operational reporting and real-time driver monitoring functions into account. Research focuses on creating an IoT-based mobile application which bus drivers will use as their main interface while bus owners will use it as their secondary interface. Users of the proposed component can manage bus availability while reporting trip progress and indicating breakdowns and obtaining operational summaries. The system utilizes onboard IoT sensor data including longitude and speed and acceleration and gyroscope readings and timestamp data to identify unsafe or inefficient driving behavior which triggers alerts through the mobile application. The component helps the smart bus management system achieve better operational understanding and driver compliance and service dependability.

1.2 Literature Review

Recent studies and research in the domain of Intelligent Transportation Systems (ITS) emphasize the increased use of Internet of Things (IoT) technologies, mobile applications, and various monitoring systems to ensure the efficient operation of vehicles and transportation systems. Most of the modern transport monitoring systems incorporate the use of Global Positioning System (GPS) technology to ensure the real-time tracking of the location of the vehicles. These systems enable transport operators to obtain various data related to the operations of the vehicles, as well as to offer real-time information related to the movement of the vehicles and the provision of transport services.

Intelligent bus monitoring systems have been developed worldwide to ensure the efficient operation of transport systems. For instance, the iBus system, implemented in London, incorporates GPS, gyroscope, and various data related to the vehicles to ensure the efficient tracking of the location of the vehicles and to offer accurate monitoring of the operations of the vehicles. In addition to this, recent research has also been conducted to monitor driver behavior with the help of sensor-based technologies. Research has proven that sensor data can be utilized to determine driver behavior, including sudden braking, aggressive acceleration, and turning behavior. These systems are designed to analyze sensor data and classify driver behavior patterns to improve road safety, vehicle efficiency, and operational monitoring. In many cases, smartphone-based systems have been utilized to effectively collect sensor data and monitor driver behavior.

Mobile devices play an important role in modern transport monitoring systems, as smartphones are equipped with various sensor technologies such as accelerometers, gyroscopes, and GPS modules that can be utilized to monitor driver behavior and improve safety. Research has been conducted to determine the capabilities of smartphone-based driver monitoring platforms, which have proven to improve driver safety with the help of driver feedback mechanisms. It has been demonstrated that mobile applications can be utilized to improve driver monitoring systems. Furthermore, IoT-based smart transportation systems often incorporate various types of sensors and onboard controllers to monitor the conditions of the vehicles and the behavior of the drivers. For instance, various studies have proposed IoT-based bus monitoring systems, incorporating accelerometer and GPS sensors to detect driving events and send the operation status to the central monitoring system. These systems highlight the capabilities of IoT technologies to enable the use of real-time monitoring, operation reporting, and safety management for transportation systems. Although various types of advancements have been achieved for the tracking of vehicles and the use of various types of sensors for transportation systems, the current systems often rely on tracking the location of the vehicles or providing various types of passenger information. Only a limited number of systems incorporate operation reporting, driver behavior monitoring, and real-time feedback using a mobile interface for transportation stakeholders. In particular, the use of IoT sensors for operation reporting and driver behavior monitoring using mobile applications needs to be developed.

Hence, it is clear that the research component of this proposed research introduces an IoT-based mobile application for bus drivers and owners. The mobile application will allow users to update bus availability information, update trip information, and notify bus breakdown information. Moreover, the analysis of sensor information such as longitude, speed, acceleration, gyroscope, and timestamp information from the IoT device will be used to identify driving activities such as overspeed driving, harsh braking, harsh turning, and slow movement. Based on these identified driving activities, alert and warning notifications will be generated to increase drivers' awareness through the mobile application. This research component will help develop an intelligent driver monitoring system for a smart bus management system.

1.3 Research Gap

While a number of Intelligent Transportation System (ITS) solutions and IoT-based bus monitoring systems have been proposed, these solutions are mostly focused on vehicle tracking, providing information services to passengers, and general vehicle monitoring. For instance, a number of researchers use GPS and RFID technologies to monitor the location and activities of buses and passengers. Using these technologies, stakeholders are able to monitor the real-time location of buses.

In addition, other researchers proposed an intelligent bus monitoring system that utilizes RFID, GPS, GPRS, and GIS technologies to identify the location of buses and store the data in a database for the purpose of monitoring. These solutions are mostly focused on improving the accuracy of vehicle tracking and transportation management by communicating data between the sensors and stakeholders. The recent IoT-based bus monitoring approaches have also included additional sensors to monitor vehicle parameters such as vehicle speed, fuel levels, accident occurrence, and driver alcohol intake. These approaches allow the authorities to monitor the vehicle's performance and conditions through IoT platforms. However, most of these approaches are focused on providing a platform for the management of vehicle conditions rather than providing an interactive platform for the drivers.

Even though there are technological advancements in IoT-based bus monitoring approaches, there are still limitations to these approaches. Most of these approaches are focused on providing a platform for vehicle tracking or sensor-based approaches rather than providing a platform for the integration of operational reporting with real-time driver feedback mechanisms. In particular, most approaches are not integrated with IoT sensors and mobile apps for providing real-time feedback to the bus drivers on their driving behaviors such as overspeeding, braking, and other driving patterns.

Moreover, the existing solutions may not include a mobile application for the operation of the service, where the bus driver and the bus owner can update the service availability, report the breakdown incident, and update the driving condition during the operation of the bus. Therefore, there is still a gap in the integration of the IoT driver behavior monitoring with the mobile application for the operation of the service.

Therefore, this research component addresses the gap in the existing solutions by proposing an IoT-integrated mobile application for the operation of the service, where the bus driver is considered the primary user of the application and the bus owner is considered the secondary user of the application. The proposed application integrates the longitude, speed, acceleration, gyroscope, and timestamp values of the bus for detecting the events such as overspeeding, aggressive braking, aggressive turning, and abnormal slow movement.

2. OBJECTIVE

2.1 Main Objective

The primary objective of this research component is to design an IoT-based mobile application that enables bus operational reporting and real-time driver behavior monitoring in a smart bus management system. This mobile application enables bus drivers to manage bus operational activities such as bus availability, trip reporting, and breakdown notifications. Furthermore, the application enables bus owners to view bus operational information.

The system utilizes IoT technology to incorporate various types of sensor data, including longitude, speed, acceleration, gyroscope, and timestamp, for analyzing bus driver behavior during bus operations. Based on the analysis of these types of sensor data, the system detects bus driver behavior such as overspeeding, abnormal slow speed, aggressive braking, and aggressive turning. Upon detecting such behavior, the system alerts and warns bus drivers using the mobile application in order to increase bus driver awareness and promote better bus driving behavior.

The operational reporting and driver behavior monitoring capabilities of the system help improve bus operational activities, bus driver discipline, and safe bus operations.

- **Specific:** The research will be specific to the development of a mobile application that incorporates IoT sensor technology to monitor driver behavior and improve operational reporting for bus services.
- **Measurable:** The system will be measurable to attain a high accuracy rate of 90-95% in identifying driving activities such as overspeeding, aggressive braking, aggressive turning, and abnormal slow movement by incorporating IoT sensor technology.
- **Achievable:** The proposed system will be achievable by incorporating mobile application technology, IoT sensor technology, and real-time data processing to identify driving patterns and alertness.
- **Relevant:** The proposed system will be relevant to resolving major operational issues affecting public bus services, such as the lack of driver behavior monitoring, insufficient operational communication, and a lack of real-time alert mechanisms.

- **Time-bound:** The proposed mobile application and driver monitoring system will be developed within an eight-month research period..

2.2 Specific Objectives

2.2.1 Mobile Application Development for Bus Operations

The first objective will be to design and develop a mobile application for bus drivers and bus owners to carry out operations within the context of the smart bus management system. The user of the application will be the bus driver, while the bus owner will be the secondary user of the application.

The application will be designed to allow users to log in to the application and carry out operations such as updating the availability of the bus, updating whether the bus is running on the designated route, reporting incidents such as breakdowns, and providing updates relating to the trips. The application will also be designed to allow users to view summaries relating to operations, including the total trips made in a day.

The development of the mobile application interface will be completed in the first four months of the research period.

2.2.2 IoT Sensor Data Integration

The second objective is to integrate the IoT sensor data received from the onboard device with the mobile application in order to enable real-time monitoring of bus movement and drivers' activities.

The IoT device will be installed in the bus to collect various data from the onboard device. The data to be collected will include GPS location in terms of longitude, current speed, acceleration values, gyroscope values, and timestamp information. The data will then be sent to the backend system to enable processing and analysis of the data received from the onboard device.

The integration of IoT sensor data with the mobile application will enable continuous monitoring of bus movement and drivers' activities. The development of the IoT data integration module will be completed within six months of the research project timeline.

2.2.3 Driver Behavior Detection Mechanism

The third objective is to develop a driver behavior detection mechanism that uses IoT sensor information for analyzing the behavior of drivers.

The system will analyze various parameters such as speed variation, acceleration, and gyroscope readings to identify various events such as overspeeding, aggressive braking, aggressive turning, and abnormal slow driving. Various rule-based logic and threshold-based analysis techniques will be employed for detecting these driving events.The driver behavior detection mechanism will ensure that at least 90-95% accuracy is achieved in detecting various

driving patterns. Development and testing of the driver behavior detection system will be achieved within seven months of the research period.

2.2.4 Real-Time Driver Alert and Notification System

The final objective of the project would be to create a system of real-time alerts and notifications for the drivers using the mobile application, to inform them of the detection of inefficient and unsafe driving behaviors.

The mobile application would create alerts or notifications for the drivers when the system detects events of overspeeding, harsh braking, or other inefficient driving behaviors. This would enable the drivers to become more aware of their behaviors and would encourage them to adopt more efficient and safe driving habits.

The system would also keep a record of the detected events and the operations, and this could be used for the purpose of monitoring and analysis by the stakeholders. This project of implementing the real-time alert system would be completed by the eighth month of the research period.

3. METHODOLOGY

3.1 Research Approach

This research component will utilize a Design Science Research (DSR) approach in designing, developing, and evaluating an IoT-enabled mobile application for bus driver monitoring and operation reporting. The reason for choosing this approach is that it is best used in developing a technological solution to real-world problems in bus operation management.

The research will be conducted by designing a mobile-based system that integrates IoT technology in monitoring and operation reporting to improve bus drivers' awareness. The solution will comprise a combination of monitoring technologies that utilize sensors in monitoring bus drivers' activities, integrated with a mobile application that will enable drivers to interact with the system in conducting their daily bus operation activities.

The development of this system will utilize an iterative approach in designing and evaluating the system. The initial system will be designed and tested using simulated data from the IoT device integrated in the bus. The system will then be refined and improved depending on the results obtained from testing the system. The evaluation will be conducted to check its ability to accurately identify driving patterns using the mobile application..

3.2 Technical Methods and Algorithm

The proposed system utilizes a variety of technology modules to achieve the desired functionality of the system. The modules include:.

3.2.1 Mobile Application Operational Module

The proposed system utilizes the mobile application as an interface between the driver and the smart bus management system. The driver of the bus is the primary user of the system, while the owner of the bus is the secondary user of the system.

The proposed system provides the following operational capabilities to the users:

- Update the availability of the bus
- Indicate whether the bus is on the assigned route or not
- Incident report of the breakdown of the bus

- View the operational summary of the system, such as the number of trips on a given day
- The system utilizes the REST API to communicate with the system.

3.2.2 IoT Sensor Data Collection and Integration

The system also integrates an IoT device placed inside the bus to collect real-time operational data. The IoT device collects data from various sensors such as:

- GPS module - to read the coordinates of the bus locations
- Speed data - to monitor the speed of the bus
- Accelerometer data - to read patterns of acceleration
- Gyroscope data - to read turning patterns of the bus
- Timestamp data - to record the timing of the event

The readings from the sensors are sent to the backend cloud platform using a secure communication protocol. The data is stored in the database and analyzed further.

3.2.3 Driver Behavior Detection Algorithm

A driver behavior detection algorithm is integrated into the system to analyze IoT sensor data and detect unsafe and inefficient driver behavior.

The IoT sensor data is analyzed to detect various parameters such as:

- Speed of the vehicle
- Accelerometer patterns
- Gyroscope rotation values
- Location and timestamp data

Based on the analysis of the parameters, the system detects unsafe and inefficient driving behavior such as:

- Overspeeding
- Abnormal slow movement of the vehicle
- Aggressive braking of the vehicle
- Aggressive turning of the vehicle

3.2.4 Real Time Alert Generation

If unsafe driving behavior is detected, a system of alert is triggered via a mobile application. The alert is a message to the driver about detected driving behavior. The alert also promotes safe driving habits.

Examples of alert messages are:

- Overspeed alert
- Aggressive braking alert
- Aggressive turn alert
- Slow driving alert

3.3 Tools and Platforms with Justification

Component	Specification	Justification
ESP32 Microcontroller	Dual-core 240 MHz, built-in Wi-Fi and Bluetooth, 520 KB SRAM	Selected as the central on-board processing unit due to its integrated Wi-Fi capability (eliminating the need for a separate communication module), low power consumption, sufficient processing power for real-time geofence calculations and IMU signal processing, the ESP32 supports the Arduino framework, enabling rapid prototyping.
Arduino IDE	Firmware development for ESP32	Used to write and upload code to the ESP32 device. Provides libraries and tools needed for GPS, sensors, and Wi-Fi communication.
Azure IoT Hub	IoT device communication and data ingestion	Allow ESP32 devices to securely send real-time data to the cloud. It manages communication between many IoT devices and the backend system.
Spring Boot	Backend REST API and business logic	Used to build the backend server that processes device data, manages APIs, and connects the database with the frontend applications.
PostgreSQL	Primary database	Stores system data such as bus routes, locations, drivers, and logs in a structured and reliable way.
React	Regulator web dashboard	Used to create an interactive web dashboard for monitoring bus locations, routes, and system data.

Table 1- Components & Justifications

Tool	Purpose	Justification
Git / GitHub	Version control and collaborative development	Used to track code changes and allow multiple developers to work on the project efficiently.
Postman	API endpoint testing and documentation	Helps test backend APIs and verify requests and responses during development.

Arduino Serial Monitor / Arduino Serial Plotter	Real-time IMU and GPS data debugging during firmware development	Used to view and debug sensor data such as GPS coordinates and motion readings from the ESP32 device.
Google Maps Roads API	Route polyline verification and road-snapping	Used to validate and align bus routes with real road paths for accurate route mapping.

Table 2- Tools & Justifications

3.4 Data Collection and Ethical Considerations

3.4.1 Data Collection Methods

The system collects data to assess the driver's behavior and system performance. The main source of data is real-time sensor data, which is used to evaluate the system's performance and the driver's behavior. The real-time data is collected from the IoT device installed in the bus.

The data collected by the system includes:

- GPS coordinates (longitude and latitude)
- Vehicle speed
- Acceleration values
- Gyroscope values
- Timestamp values

n addition to real-time data, simulated data is used to test the performance of the behavior detection algorithms in various driving scenarios such as overspeeding, hard braking, and slow movement scenarios.

Geographic route data is used to evaluate the bus route and movement. The data is collected from OpenStreetMap and digital mapping services.

3.4.2 Ethical Considerations

The system includes vehicle monitoring as well as driver behavior analysis. Therefore, ethical issues are addressed appropriately.

The testing of the system would be on a voluntary basis. The purpose of the vehicle monitoring system would also be made clear to the drivers. The system would not collect any personal data. Instead, anonymized data would be used to identify drivers. All data transmitted to the system from IoT devices would be encrypted. This would ensure proper communication. The

database would only be accessible to authorized personnel. The research is based on Sri Lanka Institute of Information Technology research ethics. The purpose of using the system is to enhance safe driving habits as well as operational efficiency, not to penalize.

3.5 Validation Strategy and Performance Metrics

The validation of the proposed system will be centered on the evaluation of the accuracy and effectiveness of the driver behavior detection mechanism and the mobile application functionality.

There are two main ways in which the system will be evaluated:

1. Using simulated sensor data
2. Using real sensor data

The performance of the system will be evaluated on the basis of various performance parameters, including:

- Driving Event Detection Accuracy-*The system should be able to detect various driving behaviors, including overspeeding and aggressive braking. The system should be able to achieve an accuracy of 90–95%.*
- Alert Response Accuracy-*The system should be able to generate alerts with high accuracy. The alert generation mechanism needs to be highly effective*.
- Data Transmission Reliability-*The system should ensure reliable communication between the IoT devices and the backend system, and between the backend system and mobile application. The system should be able to transmit data with high reliability.*
- Operational Reporting Accuracy -*The system should be able to generate operational reports with high accuracy. The system should be able to generate operational reports with high accuracy.*

HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed system component integrates a multi-layered architecture for IoT-cloud interaction that can support driver behavior monitoring and operational reporting within the smart bus management system. The architecture integrates various IoT devices, cloud processing services, and mobile application services to support data collection and analysis for real-time driver feedback and reporting.

The IoT device layer integrates an IoT device installed on each bus to collect operational and motion data from multiple sensors. The device integrates an ESP32 module with a GPS module and an IMU sensor. The GPS module collects real-time location data, including longitude and latitude values. The IMU sensor collects acceleration and gyroscope values representing

movement patterns of the bus. The sensors monitor the motion of the buses and collect data on speed, acceleration, turning, and timestamp values.

The communication layer transmits sensor data from the IoT device to the cloud platform. The ESP32 device transmits data packets to the cloud platform through wireless communication services such as Wi-Fi and cellular networks. The device uses secure communication protocols such as MQTT and HTTPS to transmit sensor data from the bus to the cloud platform.

The cloud processing layer is the core layer where all the major analytical and processing operations occur. The data received from the IoT devices is processed through the backend services created using the Spring Boot framework. This layer stores the sensor data in a database and applies driver behavior detection algorithms to analyze the driving behavior of the driver, such as overspeeding, aggressive braking, aggressive turning, and abnormal slow driving. When the driver's driving behavior is identified as dangerous, the backend system creates alert notifications for the driver.

The application layer is the layer where users can interact with the system's functionality through mobile applications. The mobile application for the bus driver is the primary interface where the driver can receive notifications, update the operational information, and send breakdown notifications. The mobile interface for the bus owner is the secondary interface where the owner can view the operational data, trip data, and driving behavior notifications.

With the integration of IoT devices, cloud processing services, and mobile applications, the proposed architecture provides the ability to continuously monitor the operational activity of the buses while ensuring the driving behavior is safe.

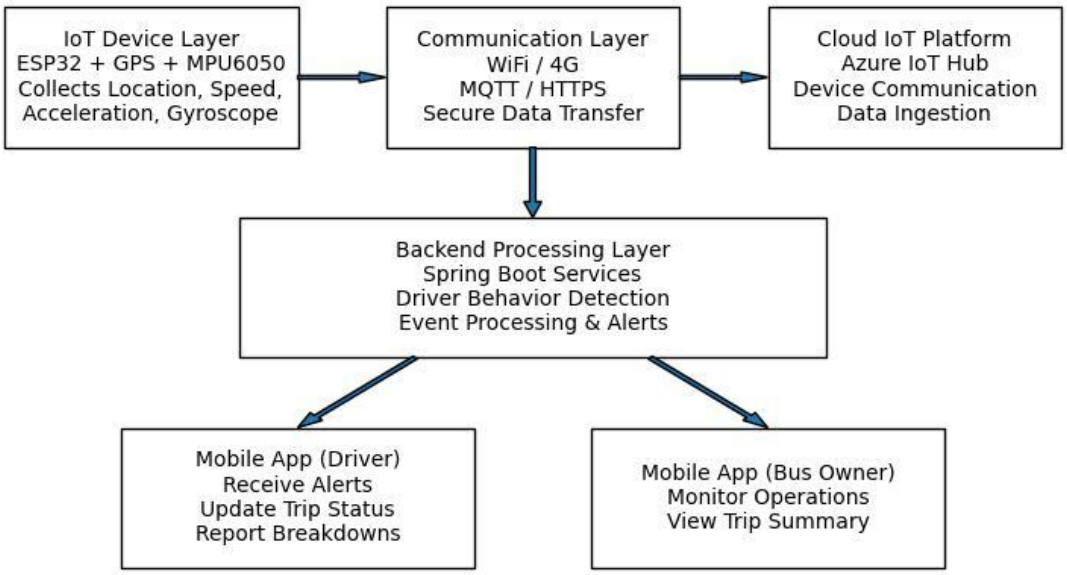


Figure 1- High-Level System Architecture

5. USER REQUIREMENTS

5.1 Stakeholder identification

Multiple stakeholders use the Policy and Time-Slot Engine system to ensure that bus operations follow approved transport policies and schedules. The primary stakeholders at transport authorities hold responsibility for creating operational policies, managing time-slot rules, and ensuring regulatory compliance. The system serves bus operators and owners who need to access service schedules and policy information that impacts their operational activities and incentive programs. Timekeepers use the system to monitor departures and verify whether buses operate within the assigned time slots. Bus drivers indirectly participate in the system because they operate according to the schedule that the policy engine produces. The stakeholders work together to establish regular bus schedules and enforce policies that lead to trustworthy public transportation services.

5.2 Functional Requirements

FR1	User Registration and Login
Input	User credentials (Route Permit No / Username, Password)
Processing	Validate credentials, encrypt password, authenticate user, and manage user session.
Output	Successful login confirmation or error message.

Table 3 - FR1

FR2	Real-Time Bus Location Tracking
Input	GPS coordinates from the bus tracking device
Processing	Receive GPS data, process location information, update system database
Output	Real-time bus location displayed on the monitoring dashboard

Table 4- FR2

FR3	Time-Slot Classification
Input	Bus trips start time, route ID, and predefined time-slot definitions (peak, normal, night).
Processing	The Policy & Time-Slot Engine compares the trip start time with configured time-slot ranges and evaluates applicable policy rules to determine the correct operational category.
Output	Assigned time-slot category for the bus operation and confirmation that the trip belongs to the defined service period.

Table 5- FR3

FR4	Trip Start and End Monitoring
Input	Trip start/end confirmation from the mobile app or GPS triggers
Processing	The system detects trip initiation and completion events and updates trip status in the database
Output	Trip status updated and displayed in the operational dashboard

Table 6- FR4

FR5	Automated Schedule Adjustment
Input	Detected service disruption events such as delayed departure, missed trip, or unavailable bus.
Processing	The system analyzes the disruption event and evaluates available time slots and policy constraints to identify the next valid operational schedule.
Output	Recommended alternative time slot or updated schedule that satisfies operational policies and maintains service continuity.

Table 7- FR5

FR6	Notification and Alert System

Input	Route deviation alerts, system status data, and high-speed
Processing	Analyze system events and trigger notifications
Output	Alerts for operators regarding route violations or high-speed

Table 8- FR6

FR7	Admin Web Dashboard
Input	Admin login credentials
Processing	Authenticate admin and allow CRUD operations
Output	Web-based dashboard with data management features

Table 9- FR7

5.3 Non-functional Requirements

NFR1	**System Reliability**
Description	The system should operate continuously with minimal downtime to ensure uninterrupted monitoring of buses.

Table 10- NFR1

NFR2	**Data Security**
Description	The system must protect sensitive information such as user credentials and financial data using secure authentication and encryption mechanisms.

Table 11- NFR2

NFR3	**Performance**
Description	The system should process GPS tracking and ticketing data quickly to provide real-time updates without delays.

Table 12- NFR3

NFR4	**Usability**
Description	The system interfaces should be easy to use for bus owners and transport authorities, with minimal training required.

Table 13- NFR4

NFR5	**Usability**
Description	The system interfaces should be easy to use for bus owners and transport authorities, with minimal training required.

Table 14- NFR5

NFR6	**Scalability**
Description	The system should support an increasing number of buses, users, and transactions as the transportation network grows.

Table 15- NFR6

6. COMMERCIALIZATION PLAN

The proposed IoT-based smart bus route monitoring system has strong potential to be developed as a practical digital solution for the public transportation sector. The system enables bus services to achieve greater operational dependability through its capabilities of monitoring routes in real time, enforcing policies automatically, and scheduling through time slots. The Policy & Time-Slot Engine component enables transport authorities to enhance their bus operation management through its ability to automate scheduling tasks that were done manually by enforcing predefined policies. Public transport management will benefit from improved operational transparency, which will decrease service interruptions and enhance decision-making capabilities.

Target Market and Customer Persona

The primary target market for this system includes government transport authorities, regional passenger transport boards, and regulatory bodies that oversee bus operations. The system will assist Sri Lankan institutions, such as the Road Passenger Transport Authority, to monitor route compliance and enforce operational policies. Private bus operators and bus owners represent secondary customers who will use the system to monitor their operational performance while ensuring they meet transport regulations.

Value Proposition

The primary value of the system brings together IoT-based monitoring systems with automatic operational management capabilities. The Policy & Time-Slot Engine enables automatic classification of service periods while it controls transport policies and makes schedule changes during service interruptions. This technology decreases the need for human monitoring operations while enhancing the reliability of bus service delivery.

Cost Estimation and Pricing Strategy

The system requires major expenses for three areas, which include IoT device installation, cloud infrastructure, software development, and system maintenance. The system operational costs will stay low because the core system functions as a software-based platform after its initial launch. The system provides two options for transportation authorities, which include a government-supported transport management platform and a licensed transport agency solution.

Revenue Model

The revenue streams for the business will derive from three different sources, which include government licensing agreements and subscription services for transport authorities and service contracts that provide bus fleet monitoring.

Competitive Advantage and Intellectual Property Considerations

The system achieves its main competitive edge through its unique capability to integrate policy-based scheduling systems with automated operational monitoring functions, which current transport monitoring systems do not offer. The research outcomes include two components, which emerge from the proposed scheduling algorithms and policy-driven operational management mechanisms that researchers can use to establish intellectual property rights.

7. BUDGET AND JUSTIFICATION

The estimated project cost includes software development, hardware infrastructure, networking services, and maintenance expenses required to deploy the proposed Smart Bus Route Monitoring and Revenue Management System. The budget reflects the costs associated with building a scalable real-time transport monitoring platform capable of supporting multiple buses and operational users.

DESCRIPTION	AMOUNT (LKR)	TOTAL (LKR)
SOFTWARE		
React.js	6 000	
Spring Boot	10 000	
React native	10 000	
C++	7 000	
PostgreSQL	9 000	
Visual Code	5 000	
Software Cost		47 000
HARDWARE		
Router	6 000	
2 pc	40 000	
IoT Device	9 000	
Total Hardware Cost		55 000
NETWORK		
Data transfer and API usage	8 000	
Hosting and server deployment	7 000	
Total Networking Cost		15 000
MAINTENCE		
Client training and technical support	6 000	
Warranty and support services	10 000	
Security software and monitoring	4 000	
System maintenance agreement	5 000	
Total Maintenance Cost		25 000
Total Project Cost		**142 000**

Table 16- Budget Calculation

8. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) provides a hierarchical representation of all tasks required for the predictive warfarin dosing system. The project breaks down into its main phases, which include requirement analysis, system design, data collection model development, implementation testing, deployment, and documentation. Each phase is further broken down into specific, manageable tasks, ensuring clarity in responsibilities and deliverables. WBS enables project teams to monitor their work as they distribute resources, manage project timelines, and identify potential risks. The project structure enables stakeholders to track project progress while upholding quality standards and verifying that the system achieves both clinical and operational goals.

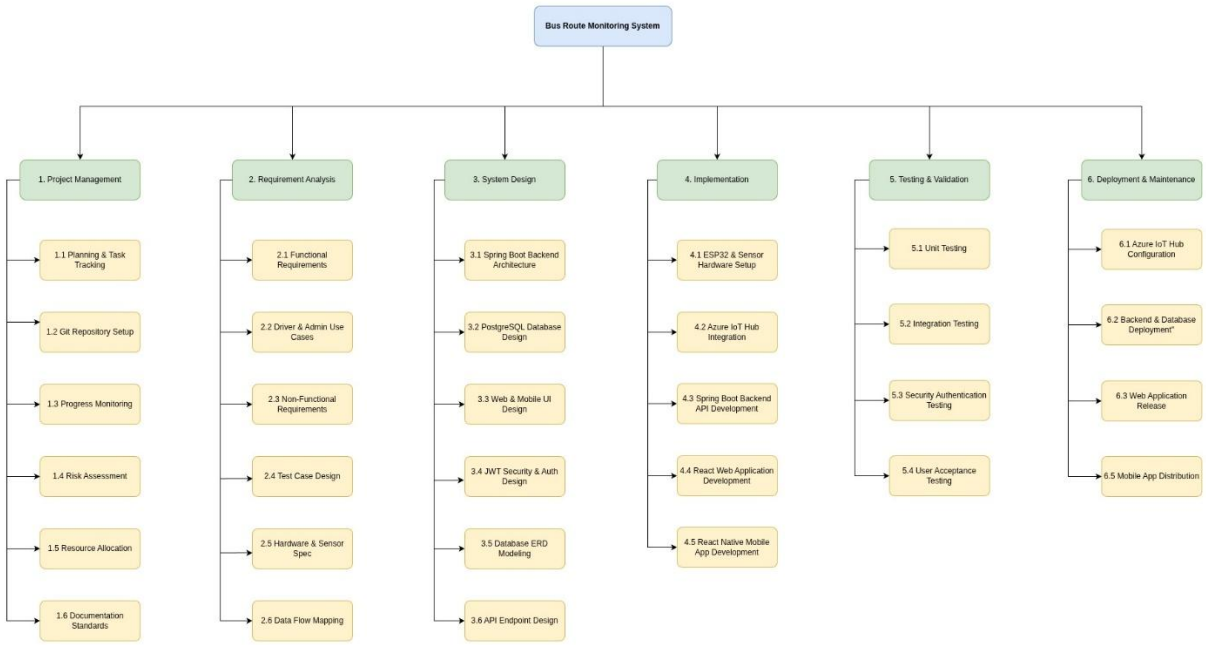


Figure 2 - Work Breakdown Chart

The Work Breakdown Structure (WBS) outlines the primary tasks necessary to complete the research project.

Task ID	Task Description	Responsible Member	Timeline
T1	Literature review and problem analysis of public bus transport monitoring systems	All Members	Weeks 1-3
T2	Collection of transport operational data, route schedules, and policy information from the Road Passenger Transport Authority	All Member	Weeks 3-5
T3	Development of an IoT-based GPS event monitoring and trip data acquisition module	Member 1	Weeks 5–7
T4	Development of route compliance verification and map-matching algorithm	Member 1	Weeks 7–10
T5	Design and implementation of Policy & Time-Slot Engine for automated scheduling and service classification	Member 2	Weeks 8–11
T6	Development of an incentive and revenue calculation module based on operational distance and service periods	Member 3	Weeks 9–12
T7	Database design and implementation for route data, policy rules, and operational logs using PostgreSQL	All Members	Weeks 10–12
T8	Development of a regulator web dashboard using React for monitoring routes and policy compliance	Members 2 & 3	Weeks 11–13
T9	Development of mobile applications for bus operators to update trips and receive operational notifications	Member 4	Weeks 11–13
T10	Integration of IoT monitoring, policy engine, revenue module, and user interfaces	All Members	Weeks 14–16
T11	System testing and validation using simulated and real operational scenarios	All Members	Weeks 16–18
T12	Performance evaluation, documentation, and final system improvements	All Members	Weeks 18–20

9. GANTT CHART

The Gantt chart shows both the research project activities schedule and their planned execution order. The document defines main project stages, which include literature review and dataset collection, model development, system implementation and testing, and final documentation. The project schedule divides tasks into multiple months to achieve continuous development and complete project milestones on time. The structured timeline enables progress monitoring of development work while ensuring both separate components and the complete system are finished during the designated research timeframe.

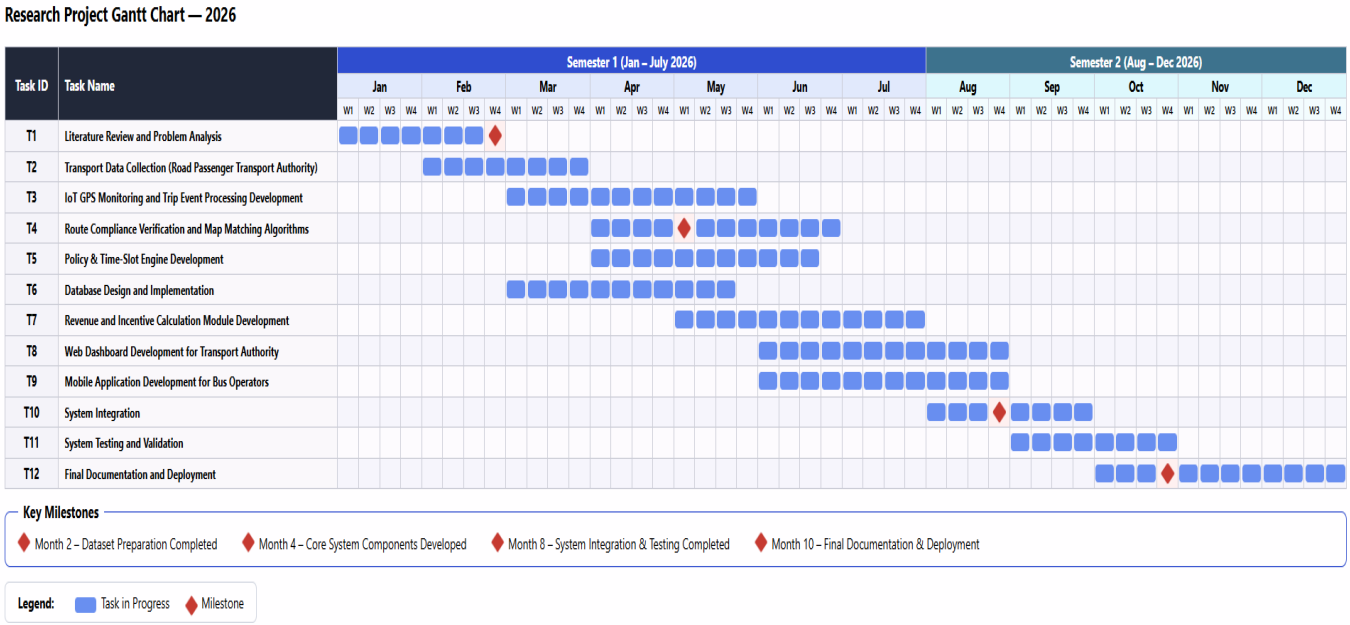


Figure 3- Gannt Chart

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APPENDICES

Appendix A – Survey / Interview Instrument

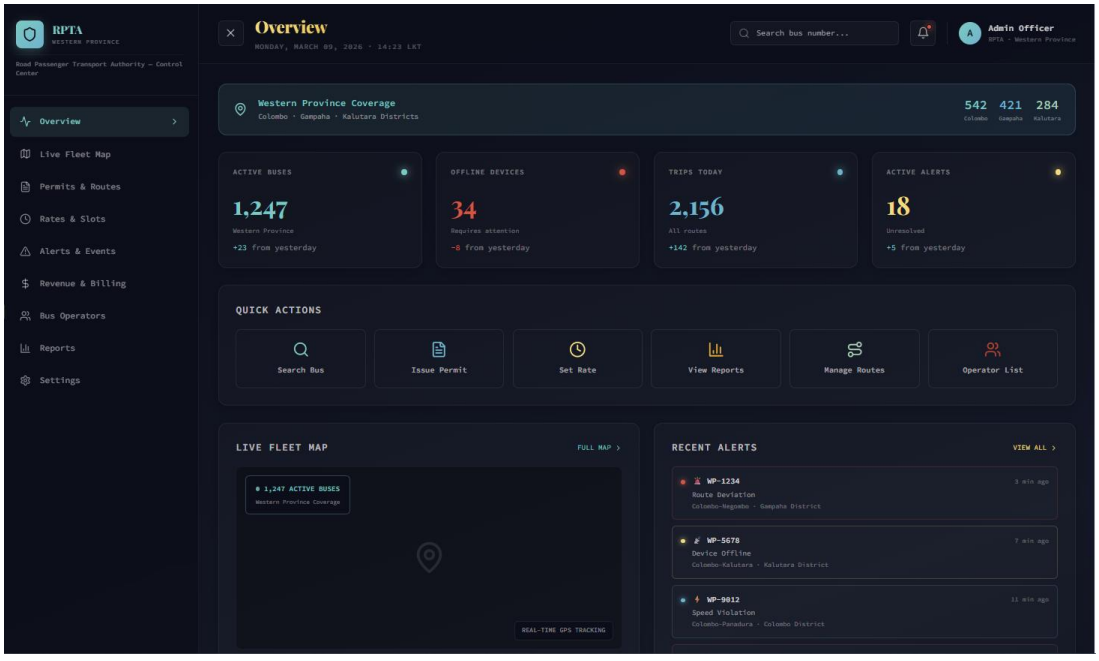
The appendix contains information collected through stakeholder discussions and consultations on public transport operations. The research data and insights were primarily derived from the Road Passenger Transport Authority, which included route tracking data, scheduling information, and details about bus operation difficulties shared by the system administrator.

Appendix B - Detailed System Diagrams

The appendix presents complete technical diagrams that were utilized throughout the system development process. The system consists of three main components, which include overall system architecture, component diagrams, and process flow diagrams to demonstrate how the IoT monitoring devices work with the cloud processing layer and Policy & Time-Slot Engine in the system.

Appendix C - Preliminary UI mockups

Sample UI for Web Application Dashboard



Appendix D - Risk Analysis

The appendix presents an overview of risks that may arise during the execution of the suggested system. The identified risks for this project include two specific hazards, which are GPS signal inaccuracies and communication interruptions that affect IoT data transmission. The system needs to face the difficulties of connecting its different elements and of collecting authentic operational data from actual environments. The document presents possible methods to reduce the identified dangers.

Appendix E - Ethical documents

Current use bus schedule

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PIL 8		**6:09**		6:24		PIL 16		12:41		12:56	
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PIL 11		**6:44**		6:59		PIL 21		13:21		13:36	
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